

Complex Analysis

5.10 Branch Point and Branch Cut

Consider a function, $\omega = z^{1/2} = f(z)$, $\omega^2 = z$, $\omega = \pm\sqrt{z}$.

1. Two copies of z -plane is required to cover $0 \rightarrow 2\pi$ in ω -plane.

- If $z = re^{i\theta}$ then $\omega = r^{1/2}e^{i\theta/2}$.
- $z = 0$, where both values are the same and both sheets touch each other is known as a Branch point.
- For $z^{\frac{1}{2}}$, there will be two sheets, these sheets known as Riemann sheet.
- For one value of z , there are two values of ω

$$\omega_1 = \sqrt{z}, \omega_2 = \sqrt{z}e^{i\pi}$$

We have to take two different values of z in-plane. 0 is the value where $\omega_1 = \omega_2$ and ∞ is the other value where $\omega_1 = \omega_2$. So 0 and ∞ are the branch points.

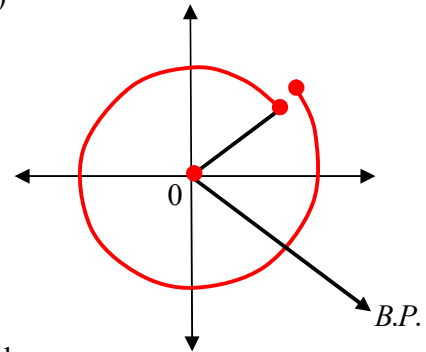
- $f(z) = (z - z_0)^{1/2} \Rightarrow$ For this z_0 and ∞ are branch points.
- $f(z) = (z - z_0)^{1/2} (z - z_{01})^{1/2} \Rightarrow$ For this z_0 & z_{01} are the branch points.
- $\therefore \frac{1}{2} + \frac{1}{2} = 1$ (Integer) $\Rightarrow \therefore \infty$ is not a branch point.

(If the sum of power is an integer then ∞ will not be Branch Point and If the sum of power is not an integer then ∞ will also be Branch Point)

$$\begin{aligned} \bullet \quad f(z) &= \log_e z, \quad z = re^{i\theta}, \quad f(z) = \log_e (re^{i\theta}) \\ f(z) &= \log_e r + i\theta = \log_e r + i(\theta + 2n\pi) \end{aligned}$$

When we take around on the plane $z = 0$, then we will not meet at the same point, we will be on the other plane.

$\log_e (z - z_0) \Rightarrow z_0$ is a branch point.



- For $n = 0$ The Riemann sheet is known as principle sheet & for $\log_e z$ there will be a branch point at $z = 0$.
- Singular points exist at $z = 0$ on all sheets of log except the principal sheet. On the principle sheet, no pole exists i.e. $z = 0$ is a removable singularity.
- $(z - z_0)^a (z - z'_0)^b$
 $\rightarrow z = z_0$ is a branch point singularity if " a " is not an integer.
 $\rightarrow z = z'_0$ is a branch point singularity if " b " is not an integer.
 $\rightarrow z = \infty$ is a branch point singularity if the summation of a & b is not an integer.

Example: Find the branch point of the following functions:

(a) $f(z) = z^{1/n}, n = 2, 3, 4, \dots$

Solution: This function has an algebraic branch point.

$$f(z) = z^{1/2}, z^{1/3}, z^{1/4}, \dots$$

$z = 0, \infty$ are the branch point.

(b) $f(z) = \log_e (1 - z)$

Solution: $z = 1$ and ∞ are the Branch point.

(c) $f(z) = \frac{1}{z} \log_e (1 - z)$

Solution: $z = 1$ & ∞ These are the branch points.

$$\frac{1}{z} \log_e(1-z) = \frac{1}{z} \left[-z - \frac{z^2}{2} - \dots \right] = -1 - \frac{z}{2}$$

$z = 0$ is a removable singularity.

(d) $f(z) = \frac{1}{z^n}, n = 1, 2, 3, \dots$

Solution: There will be no branch points.

(e) $f(z) = (z-1)^{1/2} \cdot (z-2)^{1/3}$

Solution: $1, 2, \infty$ are branch points.

$\therefore \frac{1}{2} + \frac{1}{3} \neq \text{Integer}$ so ∞ is also the branch point.

(f) $f(z) = \log_e \frac{(z-1)}{(z-2)}$

Solution: $1, 2$ are branch point.

Integrations involving a branch point:-

$$\int_0^\infty x^\alpha f(x) dx = \frac{2\pi i}{1 - e^{2\pi i \alpha}} \sum_{\text{Residue}} (z^\alpha f(z)) \quad \{\text{For non zero poles}\}$$

Example: $f(x) = \int_0^\infty \frac{x^{-\alpha}}{x+1} dx, \quad 0 < \alpha < 1$

Solution: $f(x) = \int_0^\infty \frac{x^{-\alpha}}{x+1} dx = \int_0^\infty \frac{x^{1-\alpha}}{x(x+1)} dx, \quad 0 < 1-\alpha < 1$

$$f(z) = \frac{1}{z(z+1)} \Rightarrow z = 0 \text{ \& -1}$$

$$\int_0^\infty \frac{x^{1-\alpha}}{x(x+1)} dx = \frac{2\pi i}{1 - e^{2\pi i(1-\alpha)}} \sum_{\text{Residue}} z^{1-\alpha} \cdot \frac{1}{z(z+1)} = \frac{2\pi i}{1 - e^{(1-\alpha)2\pi i}} \frac{(-1)^{1-\alpha}}{(-1)} = \frac{2\pi i(-1)^{1-\alpha-1}}{-e^{(1-\alpha)2\pi i} + 1}$$

$$\int_0^\infty \frac{x^{-\alpha}}{(x+1)} dx = \frac{2\pi i(-1)^{-\alpha}}{1 - e^{(1-\alpha)2\pi i}} = \frac{(2\pi i)e^{-i\pi\alpha}}{1 - e^{(1-\alpha)2\pi i}} = \frac{2\pi i}{1 - e^{2\pi i}} \frac{e^{-i\pi\alpha}}{e^{-2\pi i\alpha i}} = \frac{2\pi i}{e^{i\pi\alpha} - e^{-i\pi\alpha}}$$

$$\int_0^\infty \frac{x^{-\alpha}}{(x+1)} dx = \frac{2\pi i}{\sin \pi \alpha} = \frac{\pi}{\sin \pi \alpha}$$

Example: $\int_0^{\infty} \frac{x^{-1/2}}{x+1} dx$

Solution: $\int_0^{\infty} \frac{x^{-1/2}}{x+1} dx = \int_0^{\infty} \frac{x^{1/2}}{x(x+1)} dx, \quad \alpha = \frac{1}{2}$

$$\int_0^{\infty} \frac{x^{-1/2}}{x+1} dx = \frac{2\pi i}{1-e^{2\pi i/2}} \cdot \sum_{\text{Residue}} \left[z^{1/2} \cdot \frac{1}{z(z+1)} \right], \quad z=0,1$$

$$= \frac{2\pi i}{1-e^{2\pi i/2}} \cdot \frac{1}{i} = \frac{2\pi}{1-e^{\pi i}} = \frac{2\pi}{1-(-1+0)} = \pi$$

Example: $f(x) = \int_0^{\infty} \frac{[\log_e(x)]^2}{1+x^2} dx$

Solution: $f(z) = \frac{(\log_e z)^2}{1+z^2}$

$$\oint f(z) dz = 2\pi i \sum_R$$

$$\int_{-R}^{-\varepsilon} f(x) dx + \int_{C_\varepsilon} f(z) dz + \int_{\varepsilon}^R f(x) dx + \int_{C_R} f(z) dz = 2\pi i \sum_R$$

limit $R \rightarrow \infty, \varepsilon \rightarrow 0$

$$\int_{-\infty}^0 f(x) dx + 0 + \int_0^{\infty} f(x) dx + 0 = 2\pi i \sum_R$$

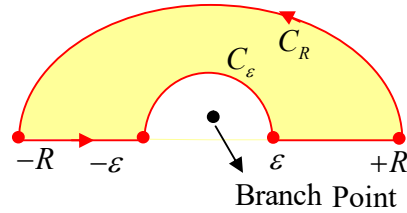
$$\int_{-\infty}^0 \frac{(\log x)^2}{1+x^2} dx + \int_0^{\infty} \frac{(\log x)^2}{1+x^2} dx = 2\pi i \sum_R$$

$$\frac{(\log z)^2}{(z+i)(z-i)}, \quad z=i$$

$$= \frac{(\log e^{i\pi/2})^2}{2i} = \frac{\left(\frac{i\pi}{2}\right)^2}{2i} = \frac{-\pi^2}{8i} \times 2\pi i = \frac{-\pi^3}{4}$$

$$-\int_0^{\infty} \frac{(\log_e x)^2}{1+x^2} dx + \int_0^{\infty} \frac{(\log_e x)^2}{1+x^2} dx = -\frac{\pi^3}{4}$$

$$-\int_0^{\infty} \frac{(\log_e(x))^2}{1+x^2} dx + \int_0^{\infty} \frac{(\log ex)^2}{1+x^2} dx = -\frac{\pi^3}{4}$$



$$\log_e(-x) = \log_e(-1) + \log_e(x) = i\pi + \log_e x$$

$$= 2 \int_0^\infty \frac{(\log x)^2}{1+x^2} dx - \pi^2 \int_0^\infty \frac{dx}{1+x^2} + 2i\pi \int_0^\infty \frac{\log_e x}{1+x^2} dx = -\frac{\pi^3}{4}$$

$$\int_0^\infty \frac{\log x}{1+x^2} dx = 0 = 2 \int_0^\infty \frac{(\log x)^2}{1+x^2} dx - \pi^2 \left[\tan^{-1} x \right]_0^\infty = -\frac{\pi^3}{4}$$

$$\int_0^\infty \frac{(\log x)^2}{1+x^2} dx = \left(-\frac{\pi^3}{4} + \frac{\pi^3}{2} \right) \left(\frac{1}{2} \right)$$

$$\int_0^\infty \frac{(\log x)^2}{1+x^2} dx = \frac{\pi^3}{8}$$

Evaluation of integration on a circular sector:

We cannot convert this integration to the range $-\infty$ to $+\infty$. However, we note that along a line

with argument $\theta = \frac{2\pi}{3}$, z^3 will have the same values as corresponding points on the real line.

$$z = re^{i\theta}, \theta = \frac{2\pi}{3}, z = re^{i2\pi/3}$$

$$z^3 = r^3 e^{2\pi i} = r^3$$

$$\oint \frac{dz}{1+z^3} = 2\pi i \sum R$$

$$\int_0^R f(x) dx + \int_{C_R} f(z) dz + \int_R^0 f(z) dz = 2\pi i \sum R$$

$$\int_0^\infty \frac{1}{1+x^3} dx + 0 + \int_\infty^0 \frac{e^{2\pi i/3} dr}{r^3 + 1} = 2\pi i \sum R \quad (dz = dr e^{i2\pi/3})$$

$$z = e^{i(2n+1)\pi/3}$$

$$z_0 = e^{i\pi/3}, z_1 = e^{i\pi}, z_2 = e^{i5\pi/3}$$

$$\frac{(1-e^{i\pi})e^{2\pi i}}{(1-e^{i\pi}) \cdot 3x^2} = \frac{1}{3} \cdot e^{-2i\pi/3} = \frac{1}{3} \left[-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right]$$

